

Operating Systems - SET 1

◆ Section 1: Booting & Basic OS Concepts

1. A system fails to move past the black screen with a blinking cursor. This usually means:

- a) The system is overheating
- b) No internet connection
- c) Bootloader isn't loading the OS
- d) Power supply is weak

Answer: c) Bootloader isn't loading the OS

Explanation: A blinking cursor on a black screen often means the system has failed to load the OS, typically due to a bootloader issue.

2. Why is an operating system needed between applications and hardware?

- a) For faster hardware installation
- b) To block access to low-level components
- c) To enable users to modify BIOS
- d) To manage interaction and resource access

Answer: d) To manage interaction and resource access

Explanation: The OS manages hardware and software resources and acts as an interface between applications and hardware.

3. A user turns on a system and hears multiple beeps before anything appears. This usually indicates:

- a) OS is updating
- b) Peripheral issue
- c) Hardware initialization error
- d) Software crash

Answer: c) Hardware initialization error

Explanation: Multiple beeps are POST (Power-On Self-Test) codes indicating hardware initialization failure.

4. When multiple applications are open, the OS decides who gets CPU time through:

- a) Disk scheduling

It is going to be hard but, hard does not mean impossible.

- b) Context filtering
- c) CPU scheduling
- d) Memory blocking

Answer: c) CPU scheduling

Explanation: CPU scheduling is used by the OS to allocate CPU time to multiple processes efficiently.

5. What does the `chmod 755 file.sh` command do?

- a) Makes the file read-only for all users
- b) Grants full permissions to everyone
- c) Grants full permissions to owner, read and execute to others
- d) Deletes the file

Answer: c) Grants full permissions to owner, read and execute to others

Explanation: `chmod 755` sets read-write-execute for owner and read-execute for group and others.

6. Which component initializes the OS at boot time?

- a) Kernel
- b) Shell
- c) Bootloader
- d) BIOS

Answer: c) Bootloader

Explanation: The bootloader is responsible for loading the OS kernel during the boot process.

7. What is the shell in an operating system?

- a) Hardware abstraction layer
- b) A text-based interface to interact with OS
- c) Memory management utility
- d) File encryption utility

Answer: b) A text-based interface to interact with OS

Explanation: The shell provides a command-line interface for users to interact with the operating system.

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◆ Section 2: Process Management & Scheduling

8. The OS maintains info about each running task. This info is stored in:

- a) Stack table
- b) Paging file
- c) Process Control Block
- d) Scheduler log

Answer: c) Process Control Block

Explanation: The Process Control Block (PCB) contains process-specific information needed by the OS.

9. When a task gets interrupted and resumes later, the system performs:

- a) Task destruction
- b) File swap
- c) Context switch
- d) User logoff

Answer: c) Context switch

Explanation: A context switch saves and loads the state of a process to resume execution.

10. When a process is ready to run but waiting for CPU, it's in the:

- a) Blocked state
- b) Waiting state
- c) Ready state
- d) Terminated state

Answer: c) Ready state

Explanation: A process in the ready state is waiting to be assigned to the processor.

11. Round Robin scheduling ensures that:

- a) Processes complete one by one in order
- b) All processes get an equal chance for CPU
- c) Longer processes run first
- d) Only priority processes run

Answer: b) All processes get an equal chance for CPU

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Explanation: Round Robin gives each process a fixed time slot in a cyclic order, promoting fairness.

12. In which case is response time prioritized over throughput?

- a) Real-time systems
- b) Batch processing
- c) Data backup
- d) Disk formatting

Answer: a) Real-time systems

Explanation: Real-time systems require fast and predictable response times.

13. Which scheduling algorithm gives the best turnaround time for short jobs?

- a) FCFS
- b) SJF
- c) RR
- d) Priority

Answer: b) SJF

Explanation: Shortest Job First (SJF) minimizes average turnaround time by executing shortest tasks first.

14. What does multithreading allow?

- a) Multiple processes running on single thread
- b) One task per process
- c) Multiple threads within a single process
- d) One thread per file

Answer: c) Multiple threads within a single process

Explanation: Multithreading enables multiple threads in the same process to run concurrently.

◆ Section 3: Memory Management

15. Paging is used to:

- a) Break files into small parts
- b) Divide memory into fixed-size units
- c) Compress data

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d) Store processes permanently

Answer: b) Divide memory into fixed-size units

Explanation: Paging divides physical memory into fixed-size blocks called pages to manage memory efficiently.

16. If memory is allocated in continuous chunks, what issue might occur?

- a) Deadlock
- b) External fragmentation
- c) Stack overflow
- d) Swapping

Answer: b) External fragmentation

Explanation: External fragmentation occurs when there's enough memory but not in contiguous blocks.

17. A process frequently swapped between RAM and disk causes:

- a) Paging boost
- b) Cache overflow
- c) Thrashing
- d) Buffering

Answer: c) Thrashing

Explanation: Thrashing happens when a process spends more time swapping than executing.

18. Which memory allocation technique may result in least waste but highest overhead?

- a) First-fit
- b) Best-fit
- c) Worst-fit
- d) Equal-fit

Answer: b) Best-fit

Explanation: Best-fit minimizes wasted space but can increase overhead due to time spent searching.

19. What is segmentation mostly based on?

- a) Time slots
- b) Process priorities

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- c) Logical divisions like code, stack, and data
- d) Disk capacity

Answer: c) Logical divisions like code, stack, and data

Explanation: Segmentation divides memory based on logical segments such as code, stack, and data.

20. **Virtual memory enables:**

- a) Running apps without CPU
- b) Use of disk as an extension of RAM
- c) Increase in CPU frequency
- d) Faster file transfer

Answer: b) Use of disk as an extension of RAM

Explanation: Virtual memory allows programs to use more memory than physically available by swapping to disk.

21. **Which table does the OS use to translate logical to physical addresses in paging?**

- a) Address bus
- b) TLB
- c) Page table
- d) Stack table

Answer: c) Page table

Explanation: The page table maps logical addresses to physical memory addresses.

♦ **Section 4: Deadlocks & Interprocess Communication**

22. When two tasks are waiting for each other's resources, we have:

- a) Multitasking
- b) Starvation
- c) Deadlock
- d) Fragmentation

Answer: c) Deadlock

Explanation: Deadlock occurs when each process in a set is waiting for another to release a resource.

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23. Deadlocks can be avoided using:

- a) File locking
- b) Wait queues
- c) Banker's algorithm
- d) Interrupts

Answer: c) Banker's algorithm

Explanation: Banker's algorithm is a deadlock-avoidance algorithm used to allocate resources safely.

24. **Interprocess communication helps multiple processes:**

- a) Use the same file only
- b) Avoid shutdowns
- c) Share data and synchronize
- d) Free memory faster

Answer: c) Share data and synchronize

Explanation: IPC allows processes to communicate and synchronize their actions.

25. **Shared memory communication requires:**

- a) A web connection
- b) Physical hardware link
- c) Memory section accessible to multiple processes
- d) Only system admin access

Answer: c) Memory section accessible to multiple processes

Explanation: Shared memory allows multiple processes to access a common memory space.

26. Semaphores are mainly used to:

- a) Format hard drives
- b) Encrypt data
- c) Prevent race conditions
- d) Handle cache misses

Answer: c) Prevent race conditions

Explanation: Semaphores are synchronization tools to prevent race conditions in concurrent systems.

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♦ **Section 5: File & Disk Management**

27. **Indexed allocation makes accessing files:**

- a) Slower than sequential
- b) Restricted to small size
- c) Flexible and direct
- d) Always read-only

Answer: c) Flexible and direct

Explanation: Indexed allocation maintains an index block pointing to file blocks for direct access.

28. **Linked file allocation causes:**

- a) Direct access
- b) Disk defragmentation
- c) Sequential access with possible delays
- d) Memory overflow

Answer: c) Sequential access with possible delays

Explanation: Linked allocation stores file blocks as a linked list, allowing sequential but not direct access.

29. **Which file allocation type best supports random access?**

- a) Contiguous
- b) Linked
- c) Indexed
- d) Flat

Answer: c) Indexed

Explanation: Indexed allocation allows random access through the index of pointers.

30. **SCAN disk scheduling is efficient because:**

- a) It moves back and forth like an elevator
- b) It deletes old files
- c) It formats disk before use
- d) It uses round-robin logic

Answer: a) It moves back and forth like an elevator

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Explanation: SCAN algorithm moves the head in one direction, serving all requests, then reverses like an elevator.

31. Which algorithm may result in starvation for outer tracks?

- a) FCFS
- b) SSTF
- c) LOOK
- d) C-SCAN

Answer: b) SSTF

Explanation: SSTF can cause starvation by favoring nearby requests and ignoring distant ones.

32. File system journaling helps in:

- a) Encrypting files
- b) Quick recovery after crash
- c) Increasing disk size
- d) Formatting file types

Answer: b) Quick recovery after crash

Explanation: Journaling keeps a log of changes, aiding recovery in case of crashes.

◆ Section 6: System Calls & Access

33. A user process makes a system call to:

- a) Bypass OS restrictions
- b) Access hardware indirectly via OS
- c) Connect to external servers
- d) Avoid security checks

Answer: b) Access hardware indirectly via OS

Explanation: System calls allow user processes to request OS services, including hardware access.

34. `fork()` in UNIX-based OS is used to:

- a) Duplicate files
- b) Switch between users
- c) Create a child process

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d) Clear memory cache

Answer: c) Create a child process

Explanation: fork() creates a new child process identical to the parent.

35. If a process replaces its code using exec(), what happens?

a) Process ends

b) It loads another program over itself

c) It enters sleep mode

d) It forks again

Answer: b) It loads another program over itself

Explanation: exec() replaces the current process's memory with a new program.

36. The wait() system call is used when:

a) You need to delay a task forever

b) A parent process is waiting for its child to end

c) A file is being downloaded

d) System is in shutdown mode

Answer: b) A parent process is waiting for its child to end

Explanation: wait() suspends execution of a parent process until its child terminates.

37. Kernel mode is required for tasks like:

a) Typing in notepad

b) Logging into Facebook

c) Managing memory or I/O

d) Changing wallpaper

Answer: c) Managing memory or I/O

Explanation: Kernel mode allows unrestricted access to system resources.

38. System calls are triggered by:

a) Clicking an icon

b) Keyboard shortcuts

c) Trap or interrupt instruction

d) BIOS reset

Answer: c) Trap or interrupt instruction

Explanation: System calls are initiated via software interrupts or trap instructions.

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39. What is the main difference between a thread and a process?

- a) Threads are slower
- b) Threads share the same memory
- c) Processes share registers only
- d) Processes can't be terminated

Answer: b) Threads share the same memory

Explanation: Threads in the same process share memory and resources, unlike processes.

40. Which of the following is not a valid IPC mechanism?

- a) Pipes
- b) Semaphores
- c) Signals
- d) Firewall

Answer: d) Firewall

Explanation: Firewall is a network security tool, not an IPC (Interprocess Communication) method.

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